ECA0806 - ANALOG AND DIGITAL COMMUNICATION

EXPERIMENT 9

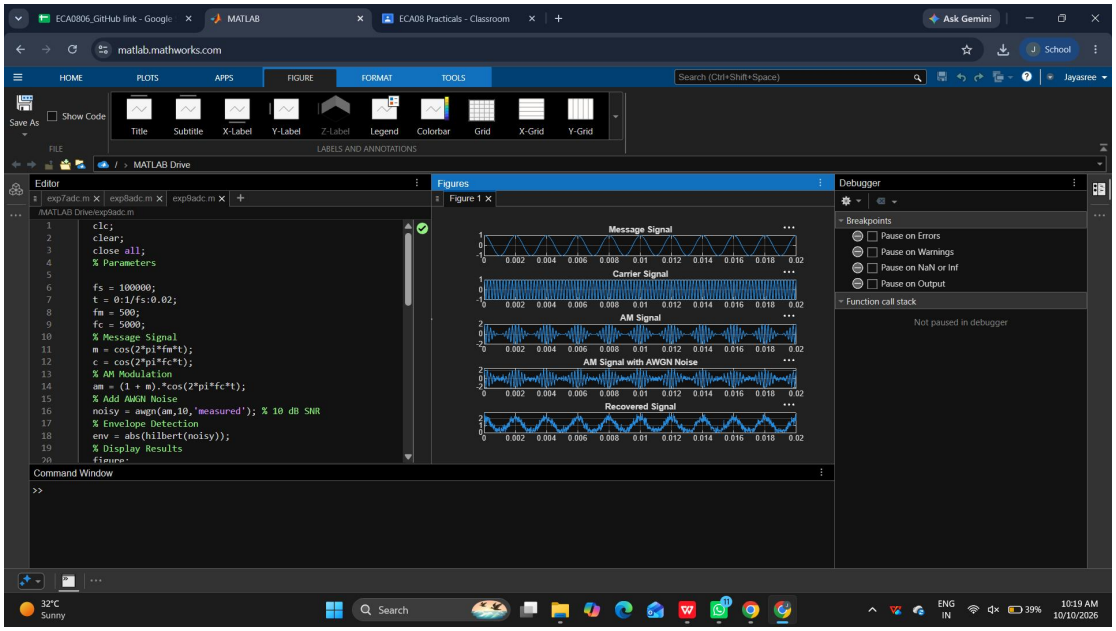
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REG NO: 192512299

PROGRAM:

```
clc;
clear;
close all;
% Parameters

fs = 100000;
t = 0:1/fs:0.02;
fm = 500;
fc = 5000;
% Message Signal
m = cos(2*pi*fm*t);
c = cos(2*pi*fc*t);
% AM Modulation
am = (1 + m).*cos(2*pi*fc*t);
% Add AWGN Noise
noisy = awgn(am,10,'measured'); % 10 dB SNR
% Envelope Detection
env = abs(hilbert(noisy));
% Display Results
figure;
subplot(5,1,1)
plot(t,m)
title('Message Signal')
grid on
subplot(5,1,2)
plot(t,c)
title('Carrier Signal')
grid on
subplot(5,1,3)
plot(t,am)
title('AM Signal')
grid on
subplot(5,1,4)
plot(t,noisy)
title('AM Signal with AWGN Noise')
grid on
subplot(5,1,5)
plot(t,env)
title('Recovered Signal')
```

grid on


ECA0806_GitHub link - Google
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Save As
Show Code
Title
Subtitle
X-Label
Y-Label
Z-Label
Legend
Colorbar
Grid
X-Grid
Y-Grid
LABELS AND ANNOTATIONS
MATLAB Drive
Editor
exp7adc.m
exp8adc.m
exp9adc.m
/MATLAB Drive/exp9adc.m
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figure;
Figures
Figure 1
Message Signal
Carrier Signal
AM Signal
AM Signal with AWGN Noise
Recovered Signal
Debugger
Breakpoints
Pause on Errors
Pause on Warnings
Pause on NaN or Inf
Pause on Output
Function call stack
Not paused in debugger
Command Window
>>
32°C
Sunny
Search
ENG
IN
39%
10:19 AM
10/10/2026